# Samar Pratap

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## Summary

Computer Science student focused on Data Engineering and DevOps. Experienced building Python/SQL ETL pipelines with PostgreSQL and FastAPI, automating deployment with Docker and GitHub Actions CI/CD, and writing tested Bash tooling for Linux automation.

## Education

**Vellore Institute of Technology** 2023 – 2027
*Bachelor of Technology – Computer Science and Engineering* *CGPA: 8.30*

## Technical Skills

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|---|---|
| **Languages** | Python, SQL, Bash |
| **Frameworks & Libraries** | FastAPI, Pandas, Pydantic, NumPy, NLTK, Matplotlib |
| **Databases** | PostgreSQL, MySQL |
| **DevOps** | Docker (+ Compose), GitHub Actions, Git, Linux |

## Project Experience

**CityAir Metrics – Real-Time Environmental Data Engineering Platform**
[github.com/Pratap-Samar/cityair-metrics]
*Python, PostgreSQL, SQL, FastAPI, Docker, GitHub Actions, pytest*

- Engineered a Python ETL pipeline integrating 2 environmental APIs across 36 cities, processing 1,872 weather and air-quality records per run with Pydantic validation, transformation, retries, deduplication, and idempotent PostgreSQL upserts.
- Built a PostgreSQL/SQL analytics layer with composite unique constraints and indexed queries to support historical time-series analysis, city trends, comparisons, and rankings, achieving sub-millisecond query latency in local benchmarks.
- Implemented a GitHub Actions CI workflow with Docker and PostgreSQL, automating code validation and database-backed checks on pushes and pull requests.

**Cue – Bash Reminder & Notification Manager**
[github.com/Pratap-Samar/cue]
*Bash, Cron, notify-send, Bats, ShellCheck, GitHub Actions*

- Developed a pure-Bash Linux reminder utility with one-time and recurring scheduling, persistent file-based storage, and desktop notifications via notify-send.
- Automated reliable reminder execution using cron and flock-protected atomic writes, supporting missed-reminder recovery, recurring interval/daily schedules, and idempotent install/uninstall workflows.
- Hardened the application with 27 automated Bats tests, ShellCheck validation, Bash syntax checks, and GitHub Actions CI, achieving 100% test-pass coverage across reminder, scheduler, concurrency, and cron workflows.

**Social Media Sentiment Analysis Dashboard**
[github.com/Pratap-Samar/Daily-Bugle-Sentiment-Dashboard]
*Python, Pandas, NLTK, PRAW, BeautifulSoup, Requests, Streamlit*

- Constructed a multi-source web scraping pipeline with BeautifulSoup and pandas to extract structured Wikipedia tables and per-page synopses across 64 storylines, using fallback parsing logic to handle inconsistent HTML structures.
- Designed an NLP preprocessing pipeline with NLTK (tokenization, stopword removal, regex normalization) and query-normalization logic to collect and clean over 7,500 Reddit comments via the PRAW API for downstream analysis.
- Applied VADER sentiment scoring to quantify fan opinion across a validated dataset of 43 story arcs, exporting results to CSV to power a reproducible, interactive Streamlit dashboard.

## Certifications

- **The Bits and Bytes of Computer Networking** – Google (Coursera) *Jul 2026*

## Co-Curricular Activities & Hobbies

- Technical Team Member, North-East Cultural Club, VIT Bhopal
- Hobbies: Reading, Cycling, Skating